

1. Replace the NaN values with correct value. And justify why you have chosen the same.

```
dataset.isna().sum()
```

```
dataset.fillna(0, inplace=True)
```

#1) Replace the missing value with central tendency-(Numeric columns → Median, Categorical columns → Mode)

#2) Entire Row delete

#3) Respective to the problem, Have to replace the missing values

#4) Consider this as a semi supervised, create a model, so that we can predict the missing values.

Here salary is sensitive columns, if not placed, definitely the salary zero, so it converts null to zero

2. How many of them are not placed?

```
dataset[dataset['status']=='Not Placed']
```

```
# How many of them are not placed? 67
```

3. Find the reason for non-placement from the dataset?

```

[5]: from scipy.stats import ttest_ind
placed= dataset[dataset['status']=='Placed']['mba_p']
not_placed= dataset[dataset['status']=='Not Placed']['mba_p']
ttest_ind(placed, not_placed)

[5]: TtestResult(statistic=np.float64(1.1259706009659818), pvalue=np.float64(0.2614450019774862), df=np.float64(213.0))

[6]: from scipy.stats import ttest_ind
placed=dataset[dataset['status']=='Placed']['etest_p']
not_placed= dataset[dataset['status']=='Not Placed']['etest_p']
ttest_ind(placed, not_placed)

[6]: TtestResult(statistic=np.float64(1.8781977946559039), pvalue=np.float64(0.061720381110174236), df=np.float64(213.0))

[7]: from scipy.stats import ttest_ind
placed=dataset[dataset['status']=='Placed']['degree_p']
not_placed= dataset[dataset['status']=='Not Placed']['degree_p']
ttest_ind(placed, not_placed)

[7]: TtestResult(statistic=np.float64(8.012038637786501), pvalue=np.float64(7.321180177806617e-14), df=np.float64(213.0))

[8]: from scipy.stats import ttest_ind
placed=dataset[dataset['status']=='Placed']['hsc_p']
not_placed= dataset[dataset['status']=='Not Placed']['hsc_p']
ttest_ind(placed, not_placed)

[8]: TtestResult(statistic=np.float64(8.266869768369684), pvalue=np.float64(1.4706851062538955e-14), df=np.float64(213.0))

[9]: from scipy.stats import ttest_ind
placed=dataset[dataset['status']=='Placed']['ssc_p']
not_placed= dataset[dataset['status']=='Not Placed']['ssc_p']
ttest_ind(placed, not_placed)

[9]: TtestResult(statistic=np.float64(11.173294337111484), pvalue=np.float64(4.115201043884438e-23), df=np.float64(213.0))

```

#overall compare mba_p, etest_p, degree_p, hsc_p, ssc_p. Based on their p values
- Placement is mainly influenced by earlier academic performance
#(10th, 12th, degree marks), while MBA performance does not show a meaningful impact. The employability test has only a weak or uncertain effect.

4. What kind of relation between salary and mba_p

4. What kind of relation between salary and mba_p

```
] dataset.iloc[:, 1:].corr(numeric_only=True)
```

```
]
          ssc_p    hsc_p  degree_p  etest_p   mba_p   salary
ssc_p  1.000000  0.513478  0.538686  0.261993  0.388478  0.558475
hsc_p  0.513478  1.000000  0.434606  0.240775  0.348452  0.459424
degree_p 0.538686  0.434606  1.000000  0.227147  0.402376  0.423762
etest_p 0.261993  0.240775  0.227147  1.000000  0.218055  0.186775
mba_p  0.388478  0.348452  0.402376  0.218055  1.000000  0.141417
salary 0.558475  0.459424  0.423762  0.186775  0.141417  1.000000
```

```
] # Salary and MBA_p correlation is 0.141417
# Here 0.1 is close to Zero and No linear relationship.
```

5. Which specialization is getting minimum salary?

```
dataset.groupby('specialisation')['salary'].mean().sort_values().head(1)
```

```
specialisation
Mkt&HR      150842.105263
Name: salary, dtype: float64
```

6. How many of them getting above 500000 salary?

```
] dataset[dataset['salary']>=500000]
```

```
]
   sl_no  gender  ssc_p  ssc_b  hsc_p  hsc_b  hsc_s  degree_p  degree_t  workex  etest_p  specialisation  mba_p  status  salary
77    78     M   64.00  Others  80.00  Others   Science    65.00   Sci&Tech   Yes    69.00     Mkt&Fin   57.65  Placed  500000.0
119   120     M   60.80  Central  68.40  Central  Commerce    64.60  Comm&Mgmt   Yes    82.66     Mkt&Fin   64.34  Placed  706250.0
150   151     M   71.00  Central  58.66  Central   Science    58.00   Sci&Tech   Yes    56.00     Mkt&Fin   61.30  Placed  690000.0
163   164     M   63.00  Others  67.00  Others   Science    64.00   Sci&Tech   No    75.00     Mkt&Fin   66.46  Placed  500000.0
174   175     M   73.24  Others  50.83  Others   Science    64.27   Sci&Tech   Yes    64.00     Mkt&Fin   66.23  Placed  500000.0
177   178     F   73.00  Central  91.15  Others  Commerce    79.00  Comm&Mgmt   Yes    89.00     Mkt&Fin   70.81  Placed  650000.0
```

7. Test the Analysis of Variance between etest_p and mba_p at significance level 5%.(Make decision using Hypothesis Testing)

```
#P value = 0.000000000000000000004672547689133748  
# it P value < 0.05, reject the null hypothesis and having significant changes
```

```

]: import scipy.stats as stats
degree = dataset[dataset["degree_t"] == 'Sci&Tech']["salary"]
specil = dataset[dataset["specialisation"] == 'Mkt&HR']["salary"]
ttest_ind(degree, specil)

]: TtestResult(statistic=np.float64(2.692041243555374), pvalue=np.float64(0.007897969943471191), df=np.float64(152.0))

]: # Pvalue = 0.007897969943471191
# Pvalue < 0.05 so the degree_t(Sci&Tech) and specialisation(Mkt&HR) having significant changes and reject null hypothesis

```

```

]: from scipy.stats import zscore
dataset['salary_zscore'] = zscore(dataset['salary'])
dataset
#here every salary zscore convert between range of small values

```

215 rows x 16 columns

10. What is the probability Density Function of the salary range from 700000 to 900000?

```
2]: mean = dataset['salary'].mean()
std= dataset['salary'].std()
from scipy.stats import norm
probability= norm.cdf(900000, mean,std)-norm.cdf(700000, mean, std)
probability
```

```
2]: np.float64(0.00041377631316941166)
```

```
4]: mean = dataset['salary'].mean()
std= dataset['salary'].std()
from scipy.stats import norm
startrangePDF = norm.pdf(700000, mean,std)
endrangePDF = norm.pdf(900000, mean,std)
print (startrangePDF,endrangePDF )
```

```
9.954477772341963e-09 4.80871130806718e-11
```

11. Test the similarity between the degree_t(Sci&Tech)with respect to etest_p and mba_p at significance level of 5%.(Make decision using Hypothesis Testing)

```
import scipy.stats as stats
degree= dataset[dataset["degree_t"]=="Sci&Tech"]["etest_p"]
specil=dataset[dataset["degree_t"]=="Sci&Tech"]["mba_p"]
ttest_ind(degree, specil)

TtestResult(statistic=np.float64(4.532000225151251), pvalue=np.float64(1.4289217003775655e-05), df=np.float64(116.0))

# P values = 0.00001429
# Here p<0.05 so degree_t(Sci&Tech)with respect to etest_p and mba_p at significance level of 5% are significantly differ and reject null hypothesis
```

12. Which parameter is highly correlated with salary?

```
] : dataset.iloc[:, 1:].corr(numeric_only=True)
# Here Salary is highly coreleated to ssc_p = 0.558475
```

```
] :
```

	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary	salary_zscore
ssc_p	1.000000	0.513478	0.538686	0.261993	0.388478	0.558475	0.558475
hsc_p	0.513478	1.000000	0.434606	0.240775	0.348452	0.459424	0.459424
degree_p	0.538686	0.434606	1.000000	0.227147	0.402376	0.423762	0.423762
etest_p	0.261993	0.240775	0.227147	1.000000	0.218055	0.186775	0.186775
mba_p	0.388478	0.348452	0.402376	0.218055	1.000000	0.141417	0.141417
salary	0.558475	0.459424	0.423762	0.186775	0.141417	1.000000	1.000000
salary_zscore	0.558475	0.459424	0.423762	0.186775	0.141417	1.000000	1.000000

13. plot any useful graph and explain it

```
] import matplotlib.pyplot as plt
placed_data = dataset[dataset['status'] == 'Placed']

plt.scatter(placed_data['etest_p'], placed_data['salary'])
plt.xlabel("Employability Test Score")
plt.ylabel("Salary")
plt.title("Salary vs Employability Test Score")
plt.show()
```

